Muhammad Junaid Jamil

66346

Lab 05

DML Command

Create Database And Table

```
USE LibraryManagementSystem;
GOCREATE TABLE Administrations (
   AdminID INT IDENTITY(1,1) PRIMARY KEY,
   AdminName VARCHAR(100) NOT NULL,
   Username VARCHAR(50) NOT NULL UNIQUE,
   Password VARCHAR(255) NOT NULL
);
CREATE TABLE Books (
   BookID INT IDENTITY(1,1) PRIMARY KEY,
   BookName VARCHAR(150) NOT NULL,
   Author VARCHAR(100),
   Publisher VARCHAR(100),
   Price DECIMAL(10,2) CHECK (Price >= 0),
   AvailableCopies INT NOT NULL DEFAULT 1
      CHECK (AvailableCopies >= 0)
);
CREATE TABLE Student (
   StudentID INT IDENTITY(1,1) PRIMARY KEY,
   StudentName VARCHAR(100) NOT NULL,
   RegistrationNo VARCHAR(30) NOT NULL UNIQUE,
   Department VARCHAR(100),
   Email VARCHAR(100) UNIQUE,
   Phone VARCHAR(20)
);
CREATE TABLE Reservation (
   ReservationID INT IDENTITY(1,1) PRIMARY KEY,
```

```
    StudentID INT NOT NULL,
    BookID INT NOT NULL,
    ReservationDate DATE NOT NULL DEFAULT GETDATE(),
    Status VARCHAR(20) NOT NULL DEFAULT 'Pending',

    FOREIGN KEY (StudentID)
        REFERENCES Student(StudentID),

    FOREIGN KEY (BookID)
        REFERENCES Books(BookID),

    CHECK (Status IN ('Pending', 'Approved', 'Cancelled'))
);
CREATE TABLE Fine (
    FineID INT IDENTITY(1,1) PRIMARY KEY,
    StudentID INT NOT NULL,
    BookID INT NOT NULL,
    FineAmount DECIMAL(10,2) NOT NULL
        CHECK (FineAmount >= 0),
    FineDate DATE NOT NULL DEFAULT GETDATE(),
    PaymentStatus VARCHAR(20) NOT NULL DEFAULT 'Unpaid',

    FOREIGN KEY (StudentID)
        REFERENCES Student(StudentID),

    FOREIGN KEY (BookID)
        REFERENCES Books(BookID),

    CHECK (PaymentStatus IN ('Unpaid', 'Paid'))
);
GO
```

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Insert records

```
INSERT INTO Administrations (AdminName, Username, Password)
VALUES
('Ali Ahmed', 'admin1', 'DemoPassword1'),
('Sara Khan', 'admin2', 'DemoPassword2');
```

INSERT INTO Books

```
(BookName, Author, Publisher, Price, AvailableCopies)
VALUES
('Introduction to SQL', 'John Smith', 'Tech Books', 1500, 5),
('Database Systems', 'Thomas Connolly', 'Pearson', 2000, 4),
('Python Programming', 'Mark Lutz', 'OReilly', 1800, 3),
('Artificial Intelligence', 'Stuart Russell', 'Pearson', 2500, 2),
('Data Warehousing', 'Paulraj Ponniah', 'Wiley', 2200, 6);
INSERT INTO Student
(StudentName, RegistrationNo, Department, Email, Phone)
VALUES
('Muhammad Ali', 'DS001', 'Data Science', 'ali@example.com', '03001234567'),
('Ayesha Khan', 'DS002', 'Computer Science', 'ayesha@example.com', '03007654321'),
('Ahmed Raza', 'DS003', 'Software Engineering', 'ahmed@example.com', '03111234567'),
('Fatima Noor', 'DS004', 'Data Science', 'fatima@example.com', '03211234567'),
('Usman Tariq', 'DS005', 'Computer Science', 'usman@example.com', '03331234567');
INSERT INTO Reservation (StudentID, BookID, ReservationDate, Status)
VALUES
(1, 1, '2026-10-01', 'Approved'),
(2, 2, '2026-10-02', 'Pending'),
(1, 3, '2026-10-03', 'Approved'),
(3, 4, '2026-10-04', 'Cancelled'),
(4, 5, '2026-10-05', 'Pending');
```

INSERT INTO Fine

```
(StudentID, BookID, FineAmount, FineDate, PaymentStatus)
```

VALUES

(1, 1, 100, '2026-10-05', 'Unpaid'),

(2, 2, 200, '2026-10-06', 'Paid'),

(3, 4, 150, '2026-10-07', 'Unpaid'),

(1, 3, 50, '2026-10-08', 'Paid'),

(4, 5, 300, '2026-10-09', 'Unpaid');

GO

Show All data

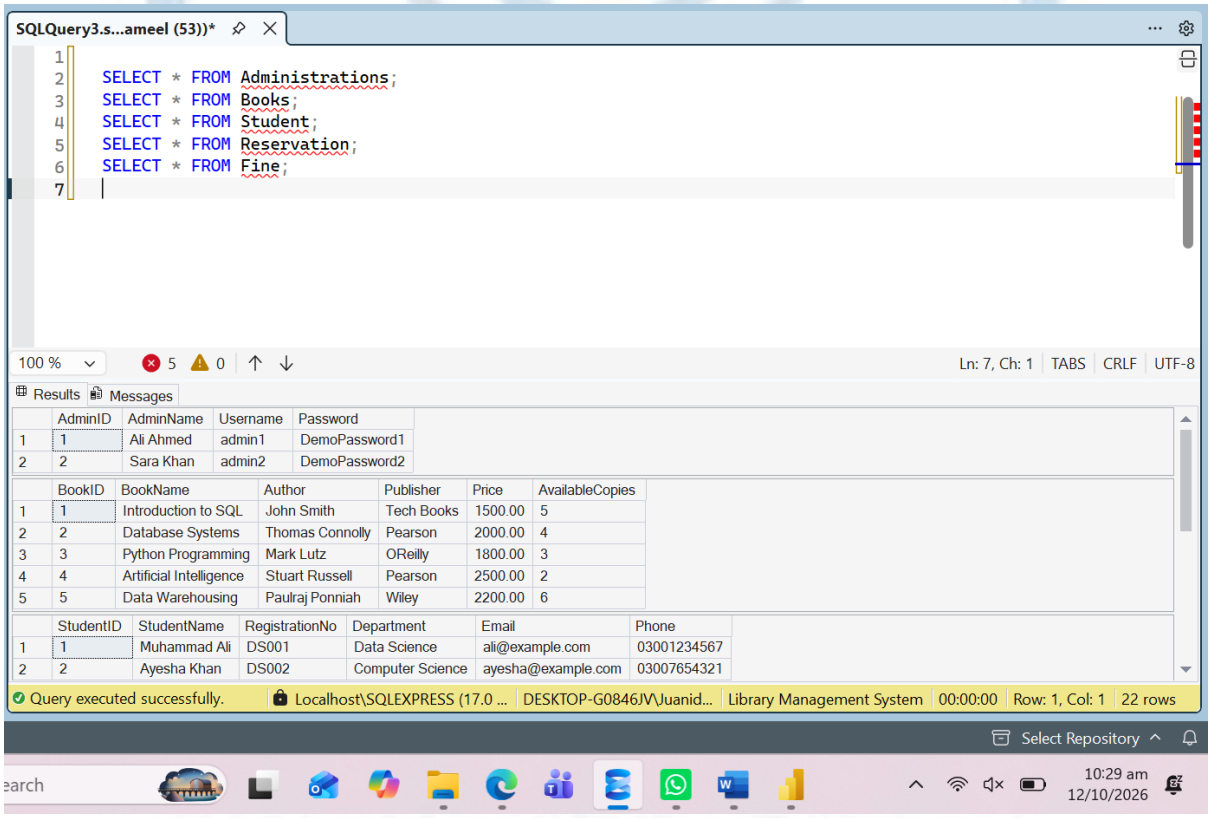
SELECT * FROM Administrations;

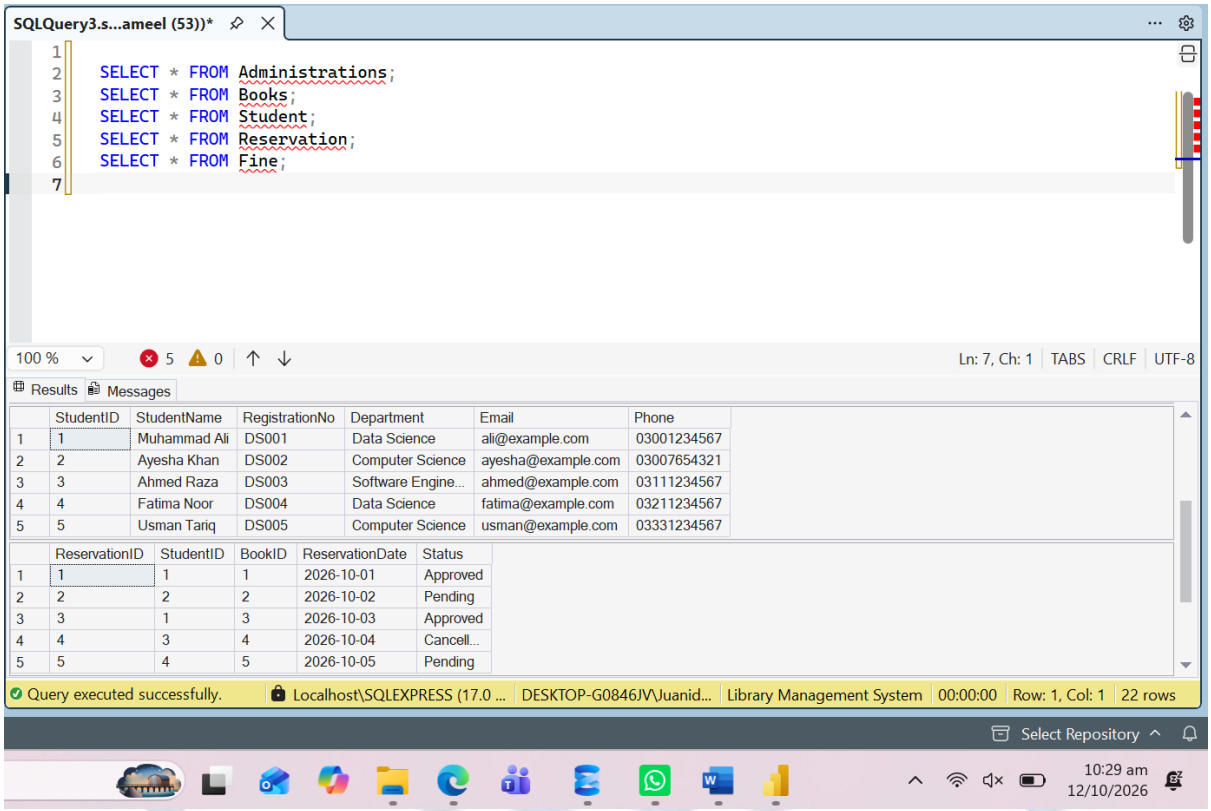
SELECT * FROM Books;

SELECT * FROM Student;

SELECT * FROM Reservation;

SELECT * FROM Fine;

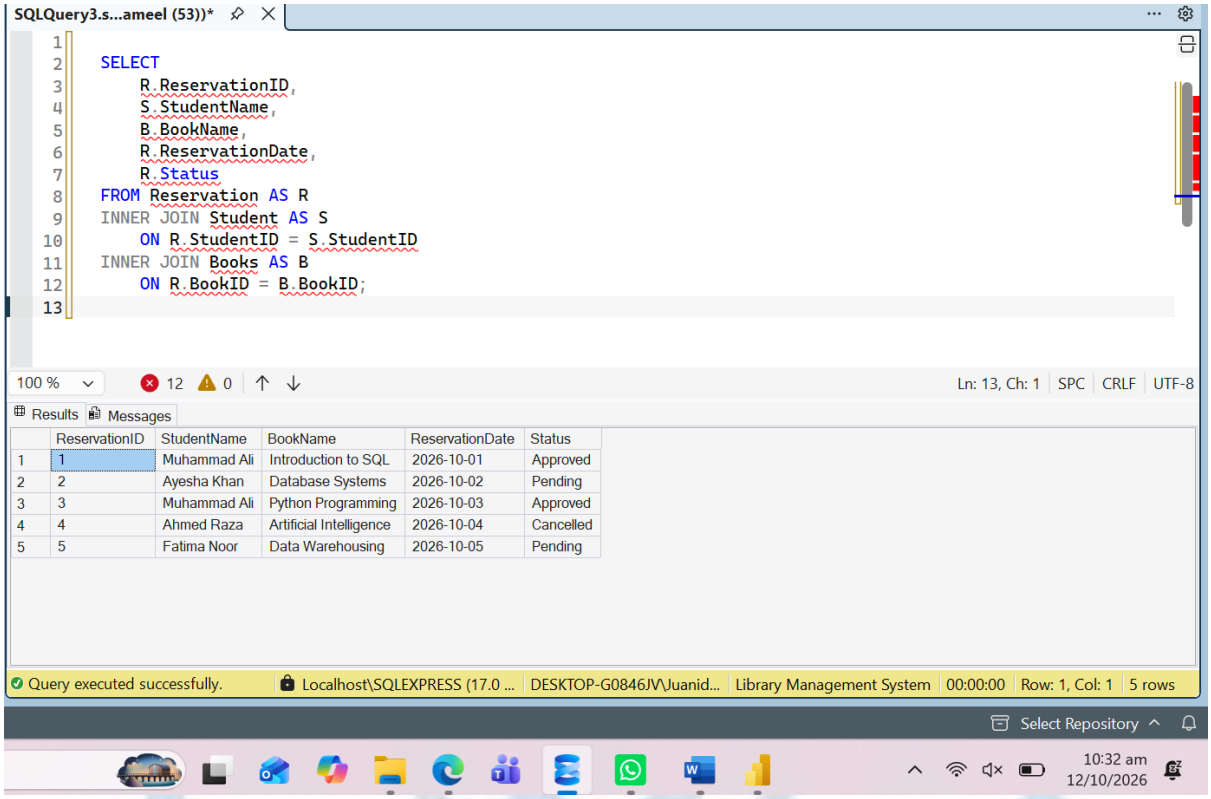




Join Queries

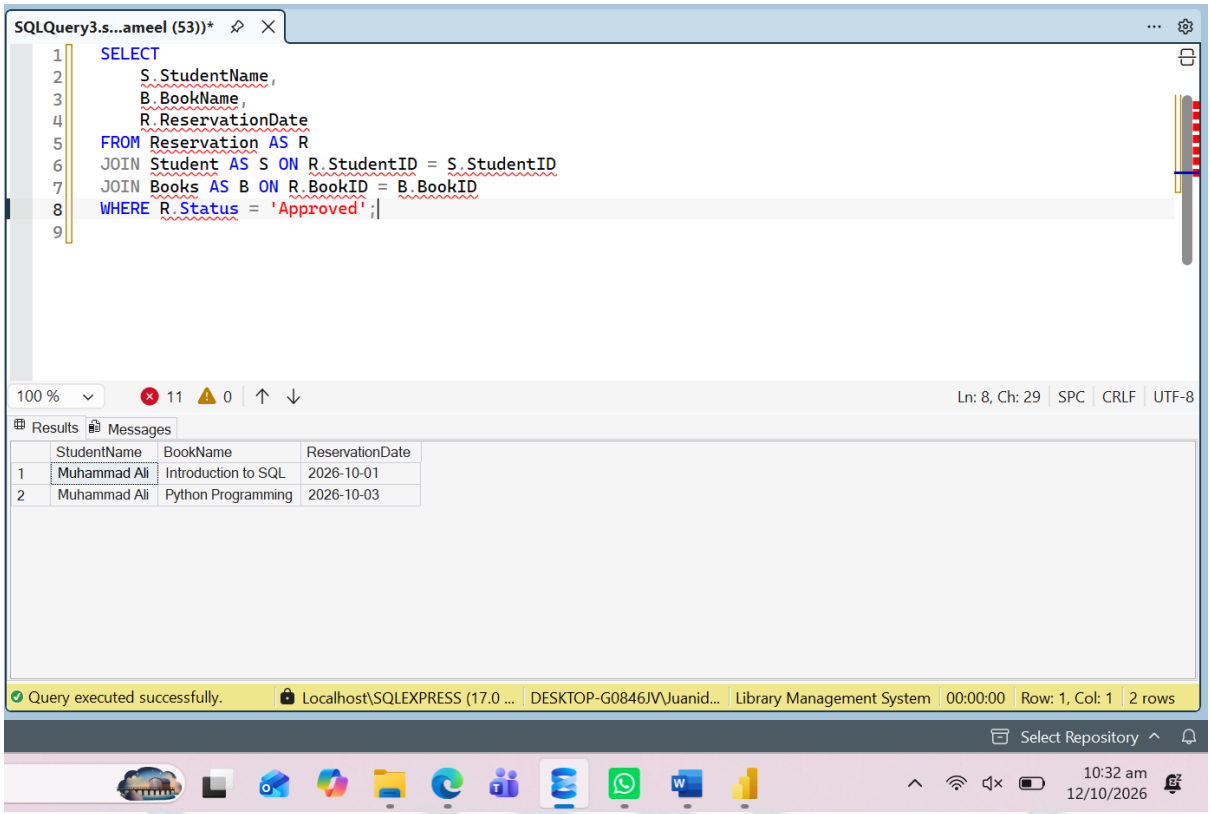
```
SELECT
    R.ReservationID,
    S.StudentName,
    B.BookName,
    R.ReservationDate,
    R.Status
FROM Reservation AS R
INNER JOIN Student AS S
    ON R.StudentID = S.StudentID
INNER JOIN Books AS B
    ON R.BookID = B.BookID;
```

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```
SELECT
    S.StudentName,
    B.BookName,
    R.ReservationDate
FROM Reservation AS R
JOIN Student AS S ON R.StudentID = S.StudentID
JOIN Books AS B ON R.BookID = B.BookID
WHERE R.Status = 'Approved';
```

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DML commands

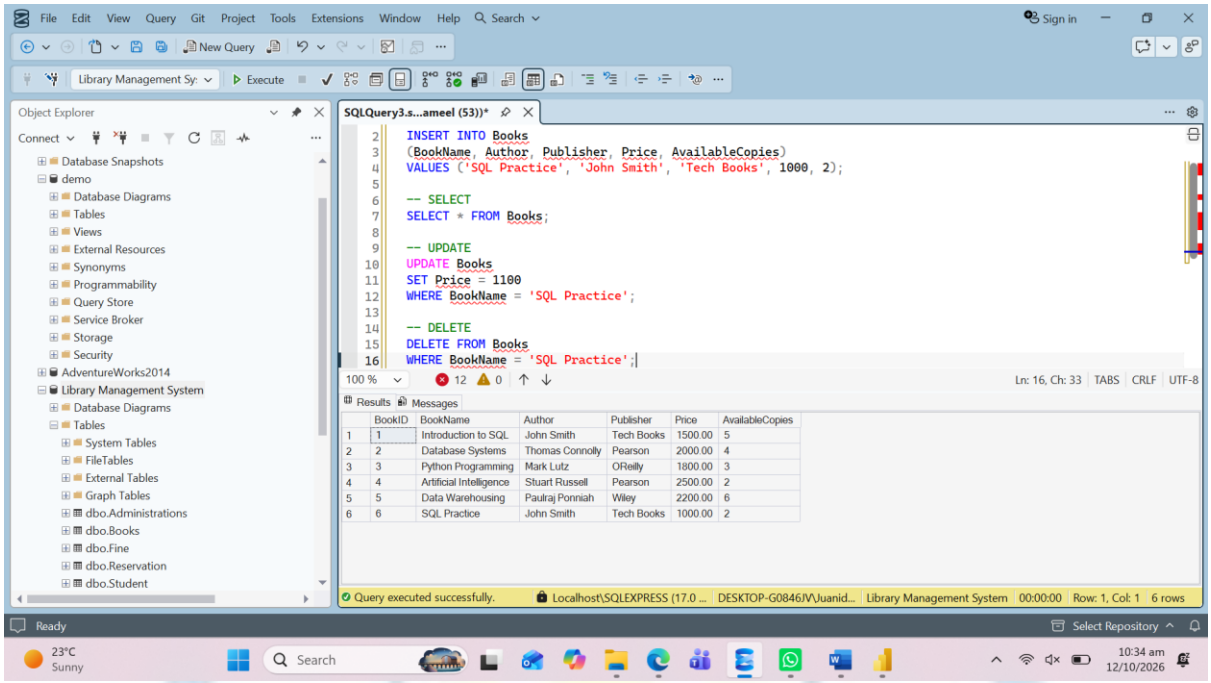
```sql
INSERT INTO Books
(BookName, Author, Publisher, Price, AvailableCopies)
VALUES ('SQL Practice', 'John Smith', 'Tech Books', 1000, 2);

-- SELECT
SELECT * FROM Books;

-- UPDATE
UPDATE Books
SET Price = 1100
WHERE BookName = 'SQL Practice';

-- DELETE
DELETE FROM Books
WHERE BookName = 'SQL Practice';
```

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DCL commands

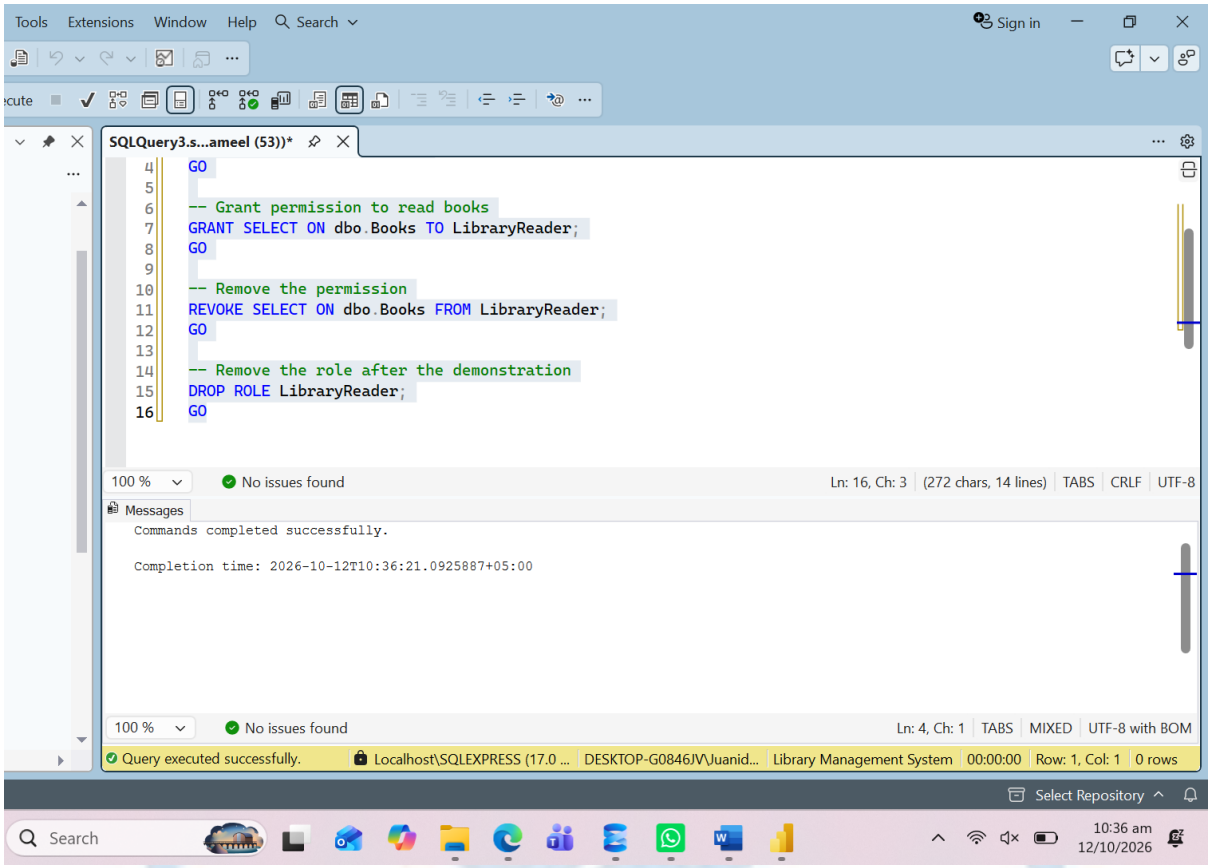
```
CREATE ROLE LibraryReader;
GO

-- Grant permission to read books
GRANT SELECT ON dbo.Books TO LibraryReader;
GO

-- Remove the permission
REVOKE SELECT ON dbo.Books FROM LibraryReader;
GO

-- Remove the role after the demonstration
DROP ROLE LibraryReader;
GO
```

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TCL commands

```
BEGIN TRANSACTION;
UPDATE Books
SET Price = Price + 100
WHERE BookID = 1;
COMMIT TRANSACTION;
SELECT * FROM Books WHERE BookID = 1;
```

